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Jessica Boyers

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Portfolio: [jessica-boyers.github.io](https://github.com/jessica-boyers)

(336)-688-5265 | Morrisville, NC

Summary: R&D technical lead with proven record of product commercialization. Master's Degree and experience in additive manufacturing, project management, and regulatory environments. Extensive experience in Design Controls and DHF creation, supported multiple 510(k) clearances. Takes ownership of outcomes and experienced in project leadership through entire product / project cycle, from start to finish.

Education: **Rochester Institute of Technology** - Rochester, NY. **August 2014 – May 2019**

GPA: 3.98/4.00 **Master of Engineering in Mechanical Engineering**, minor in Economics.

Employment:

Sovereigness Designs – Founder, Morrisville, NC.

January 2025 – Present

- Small business selling highly customizable mechatronic designs, devices, and décor at small scale.
- Designed, developed, and commercialized a mechatronic ammunition storage display product.
- Engineering Consultant – Future Hooman (Durham, NC.): Designed and fabricated mechatronic interactive art exhibits. Provided engineering guidance to artists for a variety of mechanical and electronic art pieces. Created fire safety and risk management plans, reviewed custom electronics.

Teleflex – Mechanical Engineer II, Morrisville, NC.

December 2020 – December 2024

- Technical lead responsible for entire new product development cycle of medical devices, from concept to commercialization in accordance with Design Controls and ISO13485.
- Generated multiple patents, two CAPA closures, three 510(k) clearances supported, four devices commercialized representing >\$50M revenue annually.
- Focus in concept generation, creating designs and drawings in SolidWorks, conducting tolerance analysis and simulation, evaluating DFM/DFA, and rapid prototyping and testing. Partnered with surgeons to collect and understand user needs and to evaluate prototypes to confirm performance.
- Planned and executed verification and validation testing and test method validation. Proactively derisked design and test execution. Supported Human Factors engineering activities under ANSI HE75.
- Managed multiple projects simultaneously including management of external vendors in design and manufacturing partnerships. Worked cross-functionally with RA/QA, Manufacturing and Marketing to meet project timelines and ensure Validation and Transfer success.

Energizer – Mechanical Engineer, Asheboro, NC.

June 2017 – August 2017

- Designed automated machinery improvements, streamlined changeover procedures, and determined requirements for retooling packaging lines leading to ~15% reduction in down-time of coin-cell line.

Wyman-Gordon – Mech. E. Rotational Program, Grafton, MA.

May 2016 – January 2017

- Forging process improvements and quality inspection; AS9102 certification First Article reporting.
- Led a successful 5S implementation project to develop a custom modular fixture storage solution.

Skills and Proficiencies:

Software: Microsoft Office, CAD (SolidWorks), FEA/Static simulation, MATLAB, C++, Minitab, Agile PLM.

Hardware: 3D printing; Microcontrollers; Fixturing and test equipment; Lathe, mill and laser welder.

Design: GD&T, tolerance analysis. DFM/DFA - Lean 6σ Yellow Belt. Sheet metal design, injection molding design. Project management experience, excellent verbal and written communication skills.

Medical Device: DHF creation, Design Controls, ISO 13485. Risk management, FMEA.

Patents: **US20220354484A1; US20230124563A1; Additional Application Pending Issuance**